

Full ThermoRoute architecture and calibration dataflow

Scientific modules; computational operators simplified.

(a) Inputs and masks

32-day construction buffer

WTEMP, FLOW, TEMP, PRCP, RHMEAN, DH, WDSP; each value paired with an observedness mask

Calendar and horizon

season encoding; lead $h = 1, 3, 7$ days

Forcing-regime extension (optional)

future TEMP, PRCP, RHMEAN, DH, wind F3 is a realized gridded oracle

WLEVEL excluded from every head

Predictors are dated $\leq t$; the target is never a predictor.

(c) Residual representation

Router (variable $\times$ lag)

7 variables $\times$ lags 0–14, sparsemax allocation

Left-looking temporal encoder

TCN: 2 residual blocks, kernel 3, dilations 1 and 2

Three-expert mixture

soft gate combines experts and emits the residual r

The router allocates weight over lags. It is not river-network routing; no verified topology or travel time enters.

(b) Anchor and proposal

Fixed damped-persistence branch

Climatology

c

Decay

ϕ^h

Anchor A

Learned relaxation proposal

equilibrium e and rate κ give the proposal P

κ is a statistical allocation parameter, not a heat-transfer coefficient.

(d) Heads and calibration

Combine $z = P - A + r$

Point head

$$\hat{y} = A + \delta \tanh(z/\delta), \quad \delta = 1.0 \text{ } ^\circ\text{C}$$

Quantile heads

q05, q50, q95

Event head

q90 exceedance

Calibration fitted on 2018 only

CQR offset for intervals; Platt map for event probability

Interpretive limits

$\pm 1 \text{ } ^\circ\text{C}$ bounds the deviation from the anchor only — not absolute error, event-tail error, or interval width.

F3 is a retrospective realized-meteorology oracle, not an operational forecast.

Fitting, loss, calibration and ablation detail remain in the SI text and tables.